SAFETY DATA SHEET

ATOMAL Part A

SECTION 1: IDENTIFICATION

1.1. Product identifier

Trade name: ATOMAL Part A

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture: Photographic developer
Restricted to professional users.

Uses advised against : None known.

1.3. Details of the supplier of the safety data sheet

Company and address:

ADOX Fotowerke GmbH
Pieskower Str. 30 A
15526 Bad Saarow
Deutschland
+49 (0)33631 6459-25
https://www.adox.de

E-mail: info@adox.de

SDS date: 11/22/2024

SDS Version: 1.0

1.4. Emergency telephone number

Contact the poison control at 1-800-222-1222 (24/7) or use the webPOISONCONTROL® (triage.webpoisoncontrol.org) to get specific guidance for your case
See also section 4 “First aid measures”.

SECTION 2: HAZARD(S) IDENTIFICATION

OSHA/HCS status

This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200)

2.1. Classification of the substance or mixture

Acute Tox. 4; H302, Harmful if swallowed.
Skin Corr. 1B; H314, Causes severe skin burns and eye damage.
Skin Sens. 1; H317, May cause an allergic skin reaction.
Eye Dam. 1; H318, Causes serious eye damage.
Muta. 2; H341, Suspected of causing genetic defects.
Carc. 2; H351, Suspected of causing cancer.

2.2. Label elements

Hazard pictogram(s):



Signal word: Danger

Hazard statement(s):	Harmful if swallowed. (H302) Causes severe skin burns and eye damage. (H314) May cause an allergic skin reaction. (H317) Suspected of causing genetic defects. (H341) Suspected of causing cancer. (H351)
Precautionary statement(s):	
General:	-
Prevention:	Obtain special instructions before use. (P201) Do not breathe dust. (P260) Contaminated work clothing should not be allowed out of the workplace. (P272) Wear eye protection/protective gloves/protective clothing. (P280)
Response:	IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. (P301+P330+P331) IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. (P303+P361+P353) IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. (P305+P351+P338) IF exposed or concerned: Get medical advice/attention. (P308+P313) Immediately call a POISON CENTER/doctor. (P310) If skin irritation or rash occurs: Get medical advice/attention. (P333+P313) Take off contaminated clothing and wash it before reuse. (P362+P364)
Storage:	-
Disposal:	Dispose of contents/container in accordance with local regulation (P501)
Additional labelling:	Not applicable.

2.3. Other hazards

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Not applicable. This product is a mixture.

3.2. Mixtures

Product/substance	Identifiers	% w/w	Classification	Note
1,4-dihydroxybenzene;hydroquinone;quinol	CAS No.: 123-31-9	5-10%	Acute Tox. 4, H302 Skin Sens. 1B, H317 Eye Dam. 1, H318 Muta. 2, H341 Carc. 2, H351	
4-amino-N,N-	CAS No.: 93-05-0	5-10%	Acute Tox. 3, H301	

diethylaniline;N,N-diethyl-p-phenylendiamine			Skin Corr. 1B, H314	
boric acid;boric acid;	CAS No.: 10043-35-3	3-5%	Repr. 1B, H360 (SCL: 5.50 %)	

Where the concentration of an ingredient is expressed as a range the exact concentration has been withheld as a trade secret.

See full text of H-phrases in section 16. Occupational exposure limits are listed in section 8, if these are available.

Other information

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SECTION 4: FIRST-AID MEASURES

4.1. Description of first aid measures

General information: If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Contact a doctor if in doubt about the injured person's condition or if the symptoms persist. Never give an unconscious person water or other drink.

Inhalation: Upon breathing difficulties or irritation of the respiratory tract: Bring the person into fresh air and stay with him/her.

Skin contact: Flush exposed area with water for a long time - at least 30 minutes. It may be necessary to flush for several hours. Use a comfortable water temperature (20-30 °C). Contact Poison Information/doctor/hospital for further advice on follow-up and treatment.
Remove contaminated clothing and shoes immediately. Ensure to wash exposed skin thoroughly with water and soap. Skin cleanser can be used. DO NOT use solvents or thinners.
If skin irritation occurs: Get medical advice/attention.

Eye contact: If in eyes: Flush eyes with plenty of water or salt water (20-30 °C) for at least 30 minutes and continue until irritation stops. Remove contact lenses. Make sure you flush under the upper and lower eyelids. Seek medical assistance immediately and continue flushing during transport.

Ingestion: In the case of ingestion, contact a doctor immediately. If the person is conscious, give them water. DO NOT try to induce vomiting unless this is recommended by a doctor. Hold head facing down to prevent vomit from returning to the mouth and throat. Prevent shock by keeping the injured person warm and calm. Initiate immediate resuscitation if breathing stops. If unconscious, roll the injured person into recovery position. Call an ambulance.

Burns: Not applicable.

4.2. Most important symptoms and effects, both acute and delayed

Headache, Methaemoglobinaemia (1,4-dihydroxybenzene;hydroquinone;quinol)
Sensitisation: This product contains substances, which may trigger allergic reaction upon dermal contact. Manifestation of allergic reactions typically takes place within 12-72 hours after exposure.
Tissue-damaging effects: This product contains substances with skin corrosive properties. Inhaled vapour or aerosols may produce adverse effects to lungs, irritations and burns in the respiratory organs as well as coughing. Dermal contact and contact with the eye cause irreversible effects.
High amounts of dust can cause coughing and general irritation of the respiratory airways.

4.3. Indication of any immediate medical attention and special treatment needed

IF exposed or concerned:
Get immediate medical advice/attention.

Information to medics

Bring this safety data sheet or the label from this product.

SECTION 5: FIRE-FIGHTING MEASURES

5.1. Extinguishing media

Suitable extinguishing media: Alcohol-resistant foam, carbon dioxide, powder, water mist.
Unsuitable extinguishing media: Waterjets should not be used, since they can spread the fire.

5.2. Special hazards arising from the substance or mixture

Fire will result in dense smoke. Exposure to combustion products may harm your health. Closed containers, which are exposed to fire, should be cooled with water. Do not allow fire-extinguishing water to enter the sewage system and nearby surface waters.
If the product is exposed to high temperatures, e.g. in the event of fire, dangerous decomposition compounds are produced. These are:
Sulphur oxides
Nitrogen oxides (NO_x)
Carbon oxides (CO / CO2)
Some metal oxides

5.3. Advice for firefighters

Wear self-contained breathing apparatus and protective clothing to prevent contact. Upon direct exposure contact the Poison Help Line on 1-800-222-1222 (24/7) in order to obtain further advice.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Avoid direct contact with spilled substances.
Ensure adequate ventilation, especially in confined areas.
Contaminated areas may be slippery.

6.2. Environmental precautions

Avoid discharge to lakes, streams, sewers, etc. In the event of leakage to the surroundings, contact local environmental authorities.

6.3. Methods and material for containment and cleaning up

Collect spills carefully. Moist the material with water in order to prevent the formation and propagation of dust.
Wherever possible cleaning should be performed with normal cleaning agents. Avoid use of

solvents.

6.4. Reference to other sections

See section 13 “Disposal considerations” on handling of waste.
See section 8 "Exposure controls/personal protection" for protective measures.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Avoid direct contact with the product.
Avoid contact during pregnancy and while nursing.
Smoking, drinking and consumption of food is not allowed in the work area.
See section 8 "Exposure controls/personal protection" for information on personal protection.

7.2. Conditions for safe storage, including any incompatibilities

Containers that have been opened must be carefully resealed and kept upright to prevent leakage.
Powder trickling out onto the floor or onto other containers must be prevented.

Recommended storage material:	Always store in containers of the same material as the original container.
Storage conditions:	5 - 30°C
Incompatible materials:	Acids

7.3. Specific end use(s)

This product should only be used for applications quoted in section 1.2.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

1,4-dihydroxybenzene;hydroquinone;quinol
Long term exposure limit (OSHA Table Z-1) (mg/m^3): 2
Long term exposure limit (ACGIH TLV) (mg/m^3): 1
Ceiling value (NIOSH REL) (mg/m^3): 2 [15-min]

Part 1910 - Occupational Safety and Health Standards (29 CFR 1910.1000 TABLE Z-1 - Limits for Air Contaminants)

8.2. Exposure controls

Compliance with the given occupational exposure limits values should be controlled on a regular basis.

General recommendations:	Smoking, drinking and consumption of food is not allowed in the work area.
Exposure scenarios:	There are no exposure scenarios implemented for this product.
Exposure limits:	Professional users are subjected to the legally set maximum concentrations for occupational exposure. See occupational hygiene limit values above.
Appropriate technical measures:	Do not recirculate outlet air that contain the substances. Apply standard precautions during use of the product. Avoid inhalation of gas or dust.

Airborne gas and dust concentrations must be kept at a minimum and below current limit values (see above). Installation of a local exhaust system if normal air flow in the work room is not sufficient is recommended. Ensure emergency eyewash and showers are clearly marked. Ensure that eyewash stations and safety showers are located within easy reach.
Airborne gas and dust concentrations must be kept at a minimum. Provide efficient mechanical ventilation. If not possible use suitable respiratory equipment.

Hygiene measures: In between use of the product and at the end of the working day all exposed areas of the body must be washed thoroughly. Pay special attention to hands, forearms and face.

Measures to avoid environmental exposure: Keep damming materials near the workplace. If possible, collect spillage during work.

Individual protection measures, such as personal protective equipment

Generally: Use only protective equipment with a recognized certification mark, e.g. the UL mark.

Respiratory Equipment:
No specific requirements

Skin protection:

Recommended	Type/Category	Standards	
Wear appropriate protection clothing, e.g. coveralls in polypropylene or working clothes in cotton or polyester.	-	-	

Hand protection:

Material	Glove thickness (mm)	Breakthrough time (min.)	Standards	
Nitrile	0,38	> 480	EN374-2, EN374-3, EN388	

Eye protection:

Type	Standards	
Safety glasses with side shields.	EN166	

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical state: Powder

Color: No relevant or available data due to the nature of the product.

Odor:	None
Odor threshold (ppm):	No relevant or available data due to the nature of the product.
pH:	-
pH in solution:	8 (%)
Density (g/cm³):	No relevant or available data due to the nature of the product.
Kinematic viscosity:	Does not apply to solids.
Particle characteristics:	No relevant or available data due to the nature of the product.

Phase changes

Melting point/freezing point (°F):	No relevant or available data due to the nature of the product.
Softening point/range (°F):	Does not apply to solids.
Boiling point (°F):	Does not apply to solids.
Vapor pressure:	No relevant or available data due to the nature of the product.
Relative vapor density:	Does not apply to solids.
Decomposition temperature (°F):	No relevant or available data due to the nature of the product.

Data on fire and explosion hazards

Flash point (°F):	Does not apply to solids.
Flammability (°F):	No relevant or available data due to the nature of the product.
Auto-ignition temperature (°F):	No relevant or available data due to the nature of the product.
Explosion limits (% v/v):	Does not apply to solids.

Solubility

Solubility in water:	No relevant or available data due to the nature of the product.
n-octanol/water coefficient (LogKow):	No relevant or available data due to the nature of the product.
Solubility in fat (g/L):	No relevant or available data due to the nature of the product.

9.2. Other information

Other physical and chemical parameters:	No data available.
Oxidizing properties:	No relevant or available data due to the nature of the product.

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

No data available.

10.2. Chemical stability
The product is stable under the conditions, noted in section 7 “Handling and storage”.

10.3. Possibility of hazardous reactions, including those associated with foreseeable emergencies
None known.

10.4. Conditions to avoid
None known.

10.5. Incompatible materials
Acids

10.6. Hazardous decomposition products
Under normal conditions of storage and use, hazardous decomposition products should not be produced

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on toxicological effects

Acute toxicity

Product/substance	1,4-dihydroxybenzene;hydroquinone;quinol
Test method:	OECD 401
Species:	Rat, female
Route of exposure:	Oral
Test:	LD50
Result:	367,3 mg/kg

Product/substance	1,4-dihydroxybenzene;hydroquinone;quinol
Species:	Rabbit, male/female
Route of exposure:	Dermal
Result:	>2000 mg/kg

Product/substance	boric acid;boric acid;
Species:	Rat, male/female
Route of exposure:	Oral
Test:	LD50
Result:	3450 mg/kg

Product/substance	boric acid;boric acid;
Species:	Rat, male/female
Route of exposure:	Inhalation
Test:	LC50
Result:	2,12 mg/L

Harmful if swallowed.

Skin corrosion/irritation

Product/substance	1,4-dihydroxybenzene;hydroquinone;quinol
Species:	Rabbit
Duration:	24 hours
Result:	No adverse effect observed (Not irritating)

Product/substance	boric acid;boric acid;
Species:	Rabbit, male/female
Description:	>2000 mg/kg

Result: No adverse effect observed (Not irritating)

Causes severe skin burns and eye damage.

Serious eye damage/irritation

Causes serious eye damage.

Respiratory sensitisation

Based on available data, the classification criteria are not met.

Skin sensitisation

Product/substance 1,4-dihydroxybenzene;hydroquinone;quinol
Test method: OECD 429

Germ cell mutagenicity

Product/substance boric acid;boric acid;
Species: Hamster
Conclusion: No adverse effect observed

Suspected of causing genetic defects.

Carcinogenicity

Suspected of causing cancer.

Reproductive toxicity

Product/substance boric acid;boric acid;
Conclusion: Adverse effect observed

STOT-single exposure

Based on available data, the classification criteria are not met.

STOT-repeated exposure

Based on available data, the classification criteria are not met.

Aspiration hazard

Based on available data, the classification criteria are not met.

Long term effects

Tissue-damaging effects: This product contains substances with skin corrosive properties. Inhaled vapour or aerosols may produce adverse effects to lungs, irritations and burns in the respiratory organs as well as coughing. Dermal contact and contact with the eye cause irreversible effects.

Other information

1,4-dihydroxybenzene;hydroquinone;quinol has been classified by IARC as a group 3 carcinogen.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Product/substance 1,4-dihydroxybenzene;hydroquinone;quinol
Test method: OECD 203
Species: Fish, Oncorhynchus mykiss
Duration: 96 hours
Test: LC50
Result: 0,638 mg/L

Product/substance boric acid;boric acid;
Species: Fish, Pimephales promelas
Duration: 96 hours

Test:	LC50
Result:	79,7 mg/L

Product/substance	boric acid;boric acid;
Species:	Daphnia, Daphnia magna
Duration:	48 hours
Result:	133 mg/L

Product/substance	boric acid;boric acid;
Test method:	OECD 201
Species:	Algae, Pseudokirchneriella subcapitata
Duration:	72 hours
Result:	52,4 mg/L

12.2. Persistence and degradability
Based on available data, the classification criteria are not met.

12.3. Bioaccumulative potential
Based on available data, the classification criteria are not met.

12.4. Mobility in soil
No data available.

12.5. Results of PBT and vPvB assessment
This mixture/product does not contain any substances known to fulfil the criteria for PBT and vPvB classification.

12.6. Other adverse effects
None known.

SECTION 13: DISPOSAL CONSIDERATIONS

RCRA Hazardous waste ("P" and "U" list) (40 CFR 261)
None of the components are listed

Specific labelling

Contaminated packing
Packaging containing residues of the product must be disposed of similarly to the product.

SECTION 14: TRANSPORT INFORMATION

	14.1 UN / ID	14.2 UN proper shipping name	14.3 Hazard class(es)	14.4 PG*	14.5 Env**	Other information:
DOT	UN3077	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.	Transport hazard class: 9 Label: 9 Classification code: M7 	III	No	Limited quantities: 5 kg Tunnel restriction code: (-) See below for additiona

	14.1 UN / ID	14.2 UN proper shipping name	14.3 Hazard class(es)	14.4 PG*	14.5 Env**	Other information:
						l informati on.
IMDG	UN3077	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.	Transport hazard class: 9 Label: 9 Classification code: M7 	III	No	Limited quantities: 5 kg EmS: F-A S-F See below for additional information.
IATA	UN3077	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.	Transport hazard class: 9 Label: 9 Classification code: M7 	III	No	See below for additional information.

* Packing group

** Environmental hazards

Additional information

This product is within scope of the regulations of transport of dangerous goods.
These substances when carried in single or combination packaging's containing a net quantity per single or inner packaging of 5 L or less for liquids or having a net mass per single or inner packaging of 5 kg or less for solids, are not subject to any other provisions of DOT/IMDG/IATA provided the packaging's meet the general specifications for packaging: Part 178 (DOT) / 4.1.1.1, 4.1.1.2, 4.1.1.4 - 4.1.1.8 (IMDG) / 5.0.2.4.1, 5.0.2.6.1.1, 5.0.2.8 (IATA).

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DOT / See § 172.101 Hazardous Materials Table for any information on special provisions, requirements, or warnings in connection with transport. See § 172.602, for instructions in writing regarding mitigation of damages in relation to incidents or accidents during transport.
IMDG / See section 3.2.1, for any information on special provisions, requirements, or warnings in connection with transport.
IATA / See Table 4.2 for any information on special provisions, requirements, or warnings in connection with transport.

14.6. Special precautions for user

Not applicable.

14.7. Transport in bulk according to IMO instruments

No data available.

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.2. U.S. Federal regulations

TSCA (the non-confidential portion):	Sodium sulphate is listed 1,4-dihydroxybenzene;hydroquinone;quinol is listed 4-amino-N,N-diethylaniline;N,N-diethyl-p-phenylendiamine is listed boric acid;boric acid; is listed
Clean Air Act:	1,4-dihydroxybenzene;hydroquinone;quinol is regulated as a hazardous air pollutant (HAPS)
EPCRA Section 302:	1,4-dihydroxybenzene;hydroquinone;quinol is regulated with a Treshold Planning Quantity (TPQ) of: 500/10000 pounds
EPCRA Section 304:	1,4-dihydroxybenzene;hydroquinone;quinol is regulated with a Reportable Quantity (RQ) of: 100 pounds
EPCRA section 313:	1,4-dihydroxybenzene;hydroquinone;quinol is listed
CERCLA:	1,4-dihydroxybenzene;hydroquinone;quinol is regulated with a Reportable Quantity (RQ) of: 100 pounds
Hazardous chemical inventory reporting:	This product is subject to Tier II reporting.

State regulations

California / Prop. 65:	None of the components are listed
Massachusetts / Right To Know Act:	Sodium sulphate is listed 1,4-dihydroxybenzene;hydroquinone;quinol is listed 4-amino-N,N-diethylaniline;N,N-diethyl-p-phenylendiamine is listed
New Jersey / Right To Know Act:	1,4-dihydroxybenzene;hydroquinone;quinol / Substance number: 1019
New York / Right To Know Act:	Sodium sulphate is listed Sodium sulphate is regulated with a Treshold Reporting Quantity (TRQ) of: 500 pounds — 1,4-dihydroxybenzene;hydroquinone;quinol is listed 1,4-dihydroxybenzene;hydroquinone;quinol is regulated with a Reportable Quantity (RQ) of: 1 pounds 1,4-dihydroxybenzene;hydroquinone;quinol is regulated with a Treshold Reporting Quantity (TRQ) of: 0 pounds 1,4-dihydroxybenzene;hydroquinone;quinol is regulated with a Treshold Planning Quantity (TPQ) of: 500*/10000 pounds *Quantity applies if the substance is present in the form of a fine powder (particle size less than 100 microns), molten or in solution, or reacts with water. — 4-amino-N,N-diethylaniline;N,N-diethyl-p-phenylendiamine is listed 4-amino-N,N-diethylaniline;N,N-diethyl-p-phenylendiamine is regulated with a Treshold Reporting Quantity (TRQ) of:

	100 pounds
	—
Pennsylvania / Right To Know Act:	Sodium sulphate is listed Sodium sulphate is hazardous to the environment (E)
	—
	1,4-dihydroxybenzene;hydroquinone;quinol is listed 1,4-dihydroxybenzene;hydroquinone;quinol is hazardous to the environment (E)
	—

15.4. Restrictions for application
Restricted to professional users.
Pregnant women and women breastfeeding must not be exposed to this product. The risk, and possible technical precautions or design of the workplace needed to eliminate exposure, must be considered.

15.5. Demands for specific education
No specific requirements.

15.6. Additional information
Not applicable.

15.7. Chemical safety assessment
No

15.8. Sources
OSHA Hazard Communication Standard (29 CFR 1910.1200)

SECTION 16: OTHER INFORMATION

Full text of H-phrases as mentioned in section 3
H301, Toxic if swallowed.
H302, Harmful if swallowed.
H314, Causes severe skin burns and eye damage.
H317, May cause an allergic skin reaction.
H318, Causes serious eye damage.
H341, Suspected of causing genetic defects.
H351, Suspected of causing cancer.
H360, May damage fertility or the unborn child.

The full text of identified uses as mentioned in section 1
None known.

Abbreviations and acronyms
ACGIH = American Conference of Governmental Industrial Hygienists
ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor
CAS = Chemical Abstracts Service
CERCLA = Comprehensive Environmental Response Compensation and Liability Act
DOT = Department of Transportation
EINECS = European Inventory of Existing Commercial chemical Substances
EPCRA = Emergency Planning and Community Right-To-Know Act

GHS = Globally Harmonized System of Classification and Labelling of Chemicals
HCIS = Hazardous Chemical Information System
HNOC = Hazards Not Otherwise Classified
IARC = International Agency for Research on Cancer
IATA = International Air Transport Association
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
NFPA = National Fire Protection Association
NIOSH = National Institute for Occupational Safety and Health
OECD = Organisation for Economic Co-operation and Development
OSHA = Occupational Safety and Health Administration
PBT = Persistent, Bioaccumulative and Toxic
RCRA = Resource Conservation and Recovery Act
RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail
RRN = REACH Registration Number
SARA = Superfund Amendments and Reauthorization Act
SCL = A specific concentration limit.
STEL = Short-term exposure limits
STOT-RE = Specific Target Organ Toxicity - Repeated Exposure
STOT-SE = Specific Target Organ Toxicity - Single Exposure
TSCA = The Toxic Substances Control Act
TWA = Time weighted average
UN = United Nations
UVBC = Unknown or variable composition, complex reaction products or of biological materials
VOC = Volatile Organic Compound
vPvB = Very Persistent and Very Bioaccumulative

Additional information

The classification of the mixture in regard of health hazards is in accordance with the calculation methods given by HCS (29 CFR 1910.1200).

The safety data sheet is validated by

KL

Other

A change (in proportion to the last essential change (first cipher in SDS version, see section 1)) is marked with a triangle.
The information in this safety data sheet applies only to this specific product (mentioned in section 1) and is not necessarily correct for use with other chemicals/products.
It is recommended to hand over this safety data sheet to the actual user of the product.
Information in this safety data sheet cannot be used as a product specification.
Country-language: US-en

SAFETY DATA SHEET

ATOMAL Part B

SECTION 1: IDENTIFICATION

1.1. Product identifier

Trade name: ATOMAL Part B

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses of the substance or mixture: Photographic developer
Restricted to professional users.

Uses advised against : None known.

1.3. Details of the supplier of the safety data sheet

Company and address:
ADOX Fotowerke GmbH
Pieskower Str. 30 A
15526 Bad Saarow
Deutschland
+49 (0)33631 6459-25
https://www.adox.de

E-mail: info@adox.de

SDS date: 11/22/2024

SDS Version: 1.0

1.4. Emergency telephone number

Contact the poison control at 1-800-222-1222 (24/7) or use the webPOISONCONTROL® (triage.webpoisoncontrol.org) to get specific guidance for your case
See also section 4 “First aid measures”.

SECTION 2: HAZARD(S) IDENTIFICATION

2.1. Classification of the substance or mixture

Not classified according to HCS (29 CFR 1910.1200)

2.2. Label elements

Hazard pictogram(s): Not applicable.

Signal word: Not applicable.

Hazard statement(s):

Precautionary statement(s):

General: -

Prevention: -

Response: -

Storage: -

Disposal: -

Additional labelling: Not applicable.

2.3. Other hazards

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Not applicable. This product is a mixture.

3.2. Mixtures

Product/substance	Identifiers	% w/w	Classification	Note
Sodium sulphite	CAS No.: 7757-83-7	80-95%		
orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous;	CAS No.: 1303-96-4	3-5%	Eye Irrit. 2, H319 Repr. 1B, H360 (SCL: 8.50 %)	

Where the concentration of an ingredient is expressed as a range the exact concentration has been withheld as a trade secret.

See full text of H-phrases in section 16. Occupational exposure limits are listed in section 8, if these are available.

Other information

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SECTION 4: FIRST-AID MEASURES

4.1. Description of first aid measures

General information: If breathing is irregular, drowsiness, loss of consciousness or cramps: Call 911 and give immediate treatment (first aid).
Contact a doctor if in doubt about the injured person's condition or if the symptoms persist. Never give an unconscious person water or other drink.

Inhalation: In case of discomfort: bring the person into fresh air.

Skin contact: Upon irritation: rinse with water. In the event of continued irritation, seek medical assistance.

Eye contact: Rinse gently with lukewarm water. Remove any contact lenses if this is easy to do. Continue rinsing. In case of persistent eye irritation or discomfort: Seek medical help.

Ingestion:	Rinse and flush mouth thoroughly and consume large quantities of water. In case of continued discomfort: seek medical assistance and bring this safety data sheet.
Burns:	Not applicable.

4.2. Most important symptoms and effects, both acute and delayed
High amounts of dust can cause coughing and general irritation of the respiratory airways.

4.3. Indication of any immediate medical attention and special treatment needed
Treat symptomatically.

Information to medics
Bring this safety data sheet or the label from this product.

SECTION 5: FIRE-FIGHTING MEASURES

5.1. Extinguishing media
Suitable extinguishing media: Alcohol-resistant foam, carbon dioxide, powder, water mist.
Unsuitable extinguishing media: Waterjets should not be used, since they can spread the fire.

5.2. Special hazards arising from the substance or mixture
Fire will result in dense smoke. Exposure to combustion products may harm your health. Closed containers, which are exposed to fire, should be cooled with water. Do not allow fire-extinguishing water to enter the sewage system and nearby surface waters.
If the product is exposed to high temperatures, e.g. in the event of fire, dangerous decomposition compounds are produced. These are:
Sulphur oxides
Nitrogen oxides (NO_x)
Carbon oxides (CO / CO2)
Some metal oxides

5.3. Advice for firefighters
No specific requirements.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures
Contaminated areas may be slippery.

6.2. Environmental precautions
Avoid discharge to lakes, streams, sewers, etc.
Keep unauthorized persons away from the spill

6.3. Methods and material for containment and cleaning up
Collect spills carefully. Moist the material with water in order to prevent the formation and propagation of dust.
Wherever possible cleaning should be performed with normal cleaning agents. Avoid use of solvents.

6.4. Reference to other sections
See section 13 “Disposal considerations” on handling of waste.
See section 8 "Exposure controls/personal protection" for protective measures.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Smoking, drinking and consumption of food is not allowed in the work area.
See section 8 "Exposure controls/personal protection" for information on personal protection.

7.2. Conditions for safe storage, including any incompatibilities

Recommended storage material: Always store in containers of the same material as the original container.

Storage conditions: 5 - 30°C

Incompatible materials: Acids

7.3. Specific end use(s)

This product should only be used for applications quoted in section 1.2.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

No substances are listed with a permissible exposure limit (ref: 29 CFR 1910.1000 TABLE Z-1)

8.2. Exposure controls

Apply general control to prevent unnecessary exposure

General recommendations: Smoking, drinking and consumption of food is not allowed in the work area.

Exposure scenarios: There are no exposure scenarios implemented for this product.

Exposure limits: Occupational exposure limits have not been defined for the substances in this product.

Appropriate technical measures: Apply standard precautions during use of the product.
Avoid inhalation of gas or dust.
Airborne gas and dust concentrations must be kept at a minimum. Provide efficient mechanical ventilation. If not possible use suitable respiratory equipment.

Hygiene measures: Wash hands after use.

Measures to avoid environmental exposure: No specific requirements.

Individual protection measures, such as personal protective equipment

Generally: Use only protective equipment with a recognized certification mark, e.g. the UL mark.

Respiratory Equipment:
No specific requirements

Skin protection:

Recommended	Type/Category	Standards	
Wear appropriate protection clothing, e.g. coveralls in polypropylene or working clothes in cotton or polyester.	-	-	

Hand protection:

Material	Glove thickness (mm)	Breakthrough time (min.)	Standards	
Nitrile	0,38	> 480	EN374-2, EN374-3, EN388	

Eye protection:

Type	Standards	
Safety glasses with side shields.	EN166	

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical state: Powder

Color: White

Odor: No relevant or available data due to the nature of the product.

Odor threshold (ppm): No relevant or available data due to the nature of the product.

pH: No relevant or available data due to the nature of the product.

Density (g/cm^3): No relevant or available data due to the nature of the product.

Kinematic viscosity: Does not apply to solids.

Particle characteristics: No relevant or available data due to the nature of the product.

Phase changes

Melting point/freezing point (°F): No relevant or available data due to the nature of the product.

Softening point/range (°F): Does not apply to solids.

Boiling point (°F): Does not apply to solids.

Vapor pressure: No relevant or available data due to the nature of the product.

Relative vapor density: Does not apply to solids.

Decomposition temperature (°F): No relevant or available data due to the nature of the product.

Data on fire and explosion hazards

Flash point (°F): Does not apply to solids.

Flammability (°F): No relevant or available data due to the nature of the product.

Auto-ignition temperature (°F): No relevant or available data due to the nature of the product.

Explosion limits (% v/v):	Does not apply to solids.

Solubility

Solubility in water:	No relevant or available data due to the nature of the product.
n-octanol/water coefficient (LogKow):	No relevant or available data due to the nature of the product.
Solubility in fat (g/L):	No relevant or available data due to the nature of the product.

9.2. Other information

Other physical and chemical parameters:	No data available.
Oxidizing properties:	No relevant or available data due to the nature of the product.

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity
No data available.

10.2. Chemical stability
The product is stable under the conditions, noted in section 7 “Handling and storage”.

10.3. Possibility of hazardous reactions, including those associated with foreseeable emergencies
None known.

10.4. Conditions to avoid
None known.

10.5. Incompatible materials
Acids

10.6. Hazardous decomposition products
Under normal conditions of storage and use, hazardous decomposition products should not be produced

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on toxicological effects

Acute toxicity

Product/substance	orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous;
Species:	Rat
Route of exposure:	Oral
Test:	LD50
Result:	>2500 mg/kg

Product/substance	orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous;
Species:	Rabbit
Route of exposure:	Dermal

Test:	LD50
Result:	>2000 mg/kg

Skin corrosion/irritation
Based on available data, the classification criteria are not met.

Serious eye damage/irritation
Based on available data, the classification criteria are not met.

Respiratory sensitisation
Based on available data, the classification criteria are not met.

Skin sensitisation
Based on available data, the classification criteria are not met.

Germ cell mutagenicity
Based on available data, the classification criteria are not met.

Carcinogenicity
Based on available data, the classification criteria are not met.

Reproductive toxicity
Based on available data, the classification criteria are not met.

STOT-single exposure
Based on available data, the classification criteria are not met.

STOT-repeated exposure
Based on available data, the classification criteria are not met.

Aspiration hazard
Based on available data, the classification criteria are not met.

Long term effects
None known.

Other information
None known.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity
No data available.

12.2. Persistence and degradability
Based on available data, the classification criteria are not met.

12.3. Bioaccumulative potential

Product/substance	orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous;
Conclusion:	Bioaccumulation is not expected

12.4. Mobility in soil
No data available.

12.5. Results of PBT and vPvB assessment
This mixture/product does not contain any substances known to fulfil the criteria for PBT and vPvB classification.

12.6. Other adverse effects
None known.

SECTION 13: DISPOSAL CONSIDERATIONS

RCRA Hazardous waste ("P" and "U" list) (40 CFR 261)

None of the components are listed

Specific labelling

Contaminated packing

Packaging containing residues of the product must be disposed of similarly to the product.

SECTION 14: TRANSPORT INFORMATION

	14.1 UN / ID	14.2 UN proper shipping name	14.3 Hazard class(es)	14.4 PG*	14.5 Env**	Other information:
DOT	-	-	-	-	-	-
IMDG	-	-	-	-	-	-
IATA	-	-	-	-	-	-

* Packing group

** Environmental hazards

Additional information

Not dangerous goods according to DOT, IATA and IMDG.

14.6. Special precautions for user

Not applicable.

14.7. Transport in bulk according to IMO instruments

No data available.

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

15.2. U.S. Federal regulations

TSCA (the non-confidential portion):	Sodium sulphite is listed orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous; is listed
Clean Air Act:	None of the components are listed
EPCRA Section 302:	None of the components are listed
EPCRA Section 304:	None of the components are listed
EPCRA section 313:	None of the components are listed
CERCLA:	None of the components are listed
Hazardous chemical inventory reporting:	This product is not subject to Tier II reporting.

State regulations

California / Prop. 65:	None of the components are listed
Massachusetts / Right To Know Act:	orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous; is listed
New Jersey / Right To Know Act:	None of the components are listed
New York / Right To Know Act:	orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous; is listed orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous; is regulated with a Treshold Reporting Quantity (TRQ) of: 500 pounds —
Pennsylvania / Right To Know Act:	orthoboric acid, sodium salt;;disodium tetraborate pentahydrate;disodium tetraborate decahydrate;;tetraboron disodium heptaoxide, hydrate;;disodium tetraborate, anhydrous; is listed —

15.4. Restrictions for application
Restricted to professional users.

15.5. Demands for specific education
No specific requirements.

15.6. Additional information
Not applicable.

15.7. Chemical safety assessment
No

15.8. Sources
OSHA Hazard Communication Standard (29 CFR 1910.1200)

SECTION 16: OTHER INFORMATION

Full text of H-phrases as mentioned in section 3
H319, Causes serious eye irritation.
H360, May damage fertility or the unborn child.

The full text of identified uses as mentioned in section 1
None known.

Abbreviations and acronyms
ACGIH = American Conference of Governmental Industrial Hygienists
ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road
ATE = Acute Toxicity Estimate
BCF = Bioconcentration Factor

CAS = Chemical Abstracts Service
CERCLA = Comprehensive Environmental Response Compensation and Liability Act
DOT = Department of Transportation
EINECS = European Inventory of Existing Commercial chemical Substances
EPCRA = Emergency Planning and Community Right-To-Know Act
GHS = Globally Harmonized System of Classification and Labelling of Chemicals
HCIS = Hazardous Chemical Information System
HNOC = Hazards Not Otherwise Classified
IARC = International Agency for Research on Cancer
IATA = International Air Transport Association
IMDG = International Maritime Dangerous Goods
LogPow = logarithm of the octanol/water partition coefficient
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)
NFPA = National Fire Protection Association
NIOSH = National Institute for Occupational Safety and Health
OECD = Organisation for Economic Co-operation and Development
OSHA = Occupational Safety and Health Administration
PBT = Persistent, Bioaccumulative and Toxic
RCRA = Resource Conservation and Recovery Act
RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail
RRN = REACH Registration Number
SARA = Superfund Amendments and Reauthorization Act
SCL = A specific concentration limit.
STEL = Short-term exposure limits
STOT-RE = Specific Target Organ Toxicity - Repeated Exposure
STOT-SE = Specific Target Organ Toxicity - Single Exposure
TSCA = The Toxic Substances Control Act
TWA = Time weighted average
UN = United Nations
UVBC = Unknown or variable composition, complex reaction products or of biological materials
VOC = Volatile Organic Compound
vPvB = Very Persistent and Very Bioaccumulative

Additional information

In accordance with HCS (29 CFR 1910.1200(g)), a safety data sheet is not required for this product. This safety data sheet has been created on a voluntary basis to distribute relevant information.

The safety data sheet is validated by

KL

Other

A change (in proportion to the last essential change (first cipher in SDS version, see section 1)) is marked with a triangle.
The information in this safety data sheet applies only to this specific product (mentioned in section 1) and is not necessarily correct for use with other chemicals/products.
It is recommended to hand over this safety data sheet to the actual user of the product.
Information in this safety data sheet cannot be used as a product specification.
Country-language: US-en